



Iranian Mathematical Society Competition

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– Day 1

P1 Let A and B be real matrices of sizes $n \times m$ and $m \times n$, respectively. If AB is idempotent and BA is diagonal, show that BA is also idempotent.

P2 Show that there does not exist a function $f : \mathbb{R} \rightarrow \mathbb{R}$ with the following property: "For every convergent sequence $\{a_i\}$ of pairwise distinct real numbers, the sequence $\{f(a_i)\}_{i=1}^{\infty}$ does not converge to any real number."

P3 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a multivariate polynomial with real coefficients such that the degree of each variable in f is at most 1. Define:

$$Q = \{(x_1, \dots, x_n) \mid x_1, \dots, x_n \in [0, 1]\}.$$

Show that there exists a point $b \in \mathbb{R}^n$ whose coordinates are either 0 or 1, such that for every $x = (x_1, \dots, x_n) \in Q$,
 $|f(b)| \geq |f(x)|$.

P4 Let $\{a_i\}_{i=1}^{\infty}$ be a sequence of nonnegative real numbers such that $\sum_{i=1}^{\infty} a_i < \infty$. Suppose that there are at most countably many subsets A of natural numbers for which $\sum_{i \in A} a_i$ is irrational. Prove that this sequence is eventually constant.

P5 Let G be a group, and let T and C be subgroups of G such that $G = CT$. Assume that T is finite, C is cyclic and $T \neq N_G(T)$. Prove that $G/Z(G)$ is finite.

P6 Define the set Δ as consisting of all points of the form (p_1, p_2, p_3) where p_i are nonnegative real numbers and $p_1 + p_2 + p_3 = 1$. Let v_1, v_2, v_3 be three vectors in $\mathbb{R}^3$, among which v_1 has the maximum Euclidean length. If a point (p_1, p_2, p_3) is chosen uniformly at random from Δ , prove that for every number c in the interval $(0, \frac{1}{4})$, we have:

$$\mathbb{P}(\|p_1 v_1 + p_2 v_2 + p_3 v_3\| \leq c \|v_1\|) \leq 4c.$$

(Here, $\mathbb{P}(A)$ denotes the probability of event A , and $\|w\|$ denotes the Euclidean length of vector w .)

– Day 2

- P7** In a gathering, there are n people. Every two people in this gathering are either friends or enemies. Suppose m is the maximum number of people in the gathering such that every pair among them are enemies. Represent each person as a point in the plane $\mathbb{R}^2$. Assume that the distance between any two friends is at most 1 meter, and the distance between any two enemies is more than 3 meters. Find the smallest value k such that all these people can be partitioned into k groups with the following properties:

The people in each group are all friends with each other and any two people from different groups are enemies.

- P8** Let G be a group and H a subgroup of G with finite index. If there exists a surjective homomorphism from G to the group $(\mathbb{Z}, +)$, then prove that there also exists a surjective homomorphism from H to $(\mathbb{Z}, +)$.

- P9** Let $f : (0, +\infty) \rightarrow (0, +\infty)$ be a function that is integrable on every bounded interval, and suppose that

$$\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = 1.$$

Prove that for every $\alpha < 1$,

$$\int_1^\infty \frac{f(x)}{x^\alpha} dx = \infty.$$

- P10** Let t and k be given natural numbers. Let $A_1, \dots, A_{t+1}$ be distinct k -element subsets of the set $\{1, 2, \dots, n\}$. Let $\mathcal{F}$ be a family of k -element subsets of $\{1, 2, \dots, n\}$ such that each member of $\mathcal{F}$ has empty intersection with at most $t-1$ of the sets $A_1, \dots, A_{t+1}$. Prove that for n sufficiently large,

$$|\mathcal{F}| \leq \binom{n}{k} - \binom{n - \lfloor \frac{(t+1)k}{t} \rfloor}{k}.$$

- P11** In a ring R , a subset A is called *vanishing* if for every sequence $(a_1, a_2, \dots)$ of elements from A , there exists a natural number n such that $a_1 a_2 \cdots a_n = 0$. Prove that in a commutative ring, the ideal generated by a vanishing set is itself a vanishing set.

- P12** Let α be an irrational real number, and define the function $g(z) = 1 + e^z + e^{\alpha z} : \mathbb{C} \rightarrow \mathbb{C}$. Prove that the closure of the set $\{\operatorname{Re}(z) : g(z) = 0\}$ is a bounded interval.

(Here, $\operatorname{Re}(z)$ denotes the real part of the complex number z .)